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‘Doing IT for the kids’: re-examining children, computers and the ‘information society’

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‘Children’ and ‘childhood’ have been long established as discursive sites through which adults can conceptualize and (re)construct past, present and future aspects of society. In this way the notion of the ‘child’ acts both as a nostalgic *and* futuristic device through which societal changes are portrayed and discussed in popular and political arenas. Over the last 30 years, the notion of the innocent child has been constantly evoked by politicians and the media in developed countries, in relation to diverse ‘issues’, from domestic violence to Third World famine. Thus notions of children and childhood have become central components to our understanding and shaping of an ever-changing world:

> Childhood – a temporary state – becomes an emblem for our anxieties about the passing of time, the destruction of historical formations, or conversely, a vehicle for our hopes for the future. The innocent child is caught somewhere over the rainbow – between nostalgia and utopian optimism, between the past and the future. (Jenkins, 1998: 5)

Recently, this emblematic role of the child has been exemplified in ongoing debates concerning the increasing role of technology in society and the perceived shift of countries such as the UK into a post-industrial era and associated ‘information age’. The notion of the ‘child computer user’, be it in terms of the omnipotent young computer ‘hacker’ or Internet-using ‘cyber-kid’, has proliferated in popular and political visions of the technology-based near future. Thus we are regularly informed by popular and academic commentators that the computer is the ‘children’s machine’ (Papert, 1993) and that children are ‘at the epicentre of the information revolution, ground zero of the digital world’ (Katz, 1997: 173). As

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Bingham et al. (1999: 655) observe, 'in that late 1990s, use of either of the words "childhood" or "Internet" is enough to signify at a stroke many of society's contemporary hopes and fears about what it means to be modern'.

Yet the discursive fusion of children and information and communications technology (ICT) has begun to be seen as problematic and dangerous by academic commentators, with an emerging literature contesting such 'stories' as fundamentally mis-representing the realities of children's engagement with ICT. Since the mid-1990s social scientists have begun to challenge the authenticity of contemporary discourses about children's use of ICT, as Holloway and Valentine (2001: 25) reason:

> Current public and policy understandings of children's use of new ICTs contain paradoxical ideas about childhood and technology. On the one hand, 'cybertopians' celebrate children's command of a technology which is assumed to be our future; on the other hand 'cybercritics' raise fears that this technology is putting children's emotional well being at risk. These contrasting interpretations are both problematic as they essentialise the category child, denying children's diversity and their status as social actors, and rest on technological determinist understandings of the inevitable impacts ICT has on society.

Such conclusions are grounded in a wealth of recent research emerging from the social studies of childhood and sociology of technology literatures – offering a stark contrast between the realities of children's ICT use with contemporary media and political portrayals of children and technology (e.g. Bingham et al., 1999; Downes, 1998; Facer et al., 2001; Furlong et al., 2000; Moran-Ellis and Cooper, 2000; Valentine and Holloway, 1999). Such research has taken great care to explore the many different ways that children use and do *not* use ICT in their lives – highlighting the variety of children's engagement with ICT as well as the complexity and context-driven nature of ICT as a social practice. In particular this body of research has largely focused on the child computer user and the Internet – the so-called 'cyberkid' debate. As Bingham et al. (1999) observe, at least in academic circles, the 'struggle around children and the Internet . . . has become the perhaps *the* privileged terrain on which the future (and therefore the present) of cyberspace is being played out'.

Yet in its attempts to 'reclaim' and recontextualize our view of children and technology from a grounded, empirical perspective, this emerging body of academic work has neglected to consider fully the wider societal significance of the 'child computer user' in the discursive construction of the information age. In particular, contemporary commentators have largely overlooked the political and economic foundations of what is a far more complex and long-established discourse than is acknowledged in the current literature. Indeed, the enduring and repetitive nature of many aspects of information technology and society has tended to be neglected by researchers in their attempts to make sense of the rapidly changing *present* of our technoculture. Such 'historical amnesia' has, for example,

been prevalent in much of the current work on children and the Internet. Yet, as earlier work by authors such as Haddon (1988) and McNeil (1991) reminds us, to overlook the 'near past' of technology is to overlook the substantive basis of its present. This article, therefore, offers a re-examination of the role of the child computer user as site of political contention, exploring how notions of children and technology have long been used to 'sell' technology to a society sometimes resistant to such change.

Understanding the political nature of the 'child computer user'

This article starts from the established, yet often overlooked, premise that the notions of the 'child' and 'childhood' are inherently political in nature. Childhood is, of course, a discursive invention of the past hundred years or so and the child has been used as a 'potent political metaphor' throughout post-war society. As Jenkins (1998: 2) reasons, childhood is by no means a space 'innocent' of political struggles – indeed the figure of the innocent child is central to almost all contemporary forms of politics:

> Th[e] dominant conception of childhood innocence presumes that children exist in a space beyond, above, outside the political; we imagine them to be non-combatants whom we protect from the harsh realities of the adult world, including the mud splattering of partisan politics. Yet, in reality, almost every major political battle of the twentieth century has been fought on the backs of our children . . . The innocent child carries the rhetorical force of such arguments; we are constantly urged to take action to protect our children.. . . We opportunistically evoke the figure of the innocent child as a 'human shield' against criticism.

In following this line of reasoning, we must also be aware of the socially constructed nature of 'technology' as a discursive devise. As Bromley (1997) argues, the ways that technologies are described shape what we do and do not see in them, and how they are ultimately used and treated in society. In examining societal discourses about children and technology it is therefore important to consider what such 'stories' omit (and therefore imply as insignificant) and question the assumptions presented to us as 'fact'. Thus, the rhetoric used to shape these notions of the 'child computer user' can help reveal the structures of political power and economic shaping concerns behind the 'information age':

> Language, written and spoken (or signed), does not simply convey *objective* statements of fact or even expressions of intent. Language also transmits various semantic codes that have underlying narratives, stories and ways of seeing the world.. . . Clearly, language is used to establish or reinforce various representations of power, and those aspects of language are of interest to understanding politics. (Sussman, 1997: 30)

From this background, this article now re-examines the contemporary meanings and functions of the 'child computer user' in terms of the conscious and unconscious role(s) it has played in British life over the last two decades; following Livingstone's (1998) broad use of the term 'child' due to 'the awkward absence of a single term to cover our age range, extended to cover 6 to 17 years'. In doing so the article draws on previous approaches to examining the social shaping of technology by focusing on three discursive arenas. First, the broadcast and print media – in the form of newspapers, television, film and radio – have long played an integral role in sensitizing the public to issues of children and childhood and technology (Berdayes and Berdayes, 1998; Scheper-Hughes and Stein, 1998). Moreover, the commercial advertising and marketing of the computer also provides a ready insight into the rationales, strategies and motivations of private industry (Cortada, 1993; Martin, 1993; Selwyn, 1999a). Finally, the article also draws upon political discourse – in the form of policy statements, official announcements and other party political dissemination (Moran-Ellis and Cooper, 2000; Selwyn, 2000).

In examining these discursive sites, the article takes a historical perspective rather than just focusing on a current issue of concern such as Internet pornography or mobile telephony. A longitudinal perspective is important in highlighting the development of 'common-sense' assumptions that inform public discourse about children (Kitzinger, 1999) as well as encompassing the rapid development of 'computer' technologies from the video game and computer programming to the world-wide web and computer-mediated communication. In this way the article focuses specifically on the last 20 years as constituting the substantive period of public information technology use in the UK. The end of the 1970s has been identified by sociologists of technology as a significant milestone in the history of computer technologies, which were still in the process of formation and had yet to be stabilized or naturalized into the daily lives of the general public (Reed, 2000). Indeed, even by 1980, the computer was by no means successfully integrated into mainstream British culture. Therefore the choice of 1980 as a starting point for our present analysis has deliberate technological, social *and* political connotations. Marking as it did the beginnings of Margaret Thatcher's Conservative government, the early commercial production in the UK and US of home computers as well as a growing media interest in new technologies, 1980 can be seen as a distinctive 'turning point' in the UK's relationship with information technology.

This article now goes onto present an analysis of the different discursive constructions of the child computer user in the UK over the last 20 years. Due to constraints of space this analysis is necessarily summary but draws upon a systematic analysis of the following sources:

- the UK parliamentary record Hansard (1981–2001);
- the NISS Clover database of newspapers and magazine titles (1980–2000);
- *The Times* index of newspaper stories (covering *The Times*, the *Sunday Times*, *The Telegraph*, *The Observer* and the *Financial Times* newspapers) (1980–1992);
- Online databases of *The Times* and *Sunday Times*, *The Guardian*, the *Daily Mail* and *Sunday Mail*, *The Observer* and the *Financial Times* newspapers (1993–2000);
- BBC News Online Archives (1997–2001);
- The Adflip online database of advertisements and commercial marketing (1980–2000);
- *Educational Computing and Technology* magazine (1982–2001).

From the analysis of these sources, six themes are identified and discussed, namely: (1) the 'natural' child computer user; (2) the 'successful' child computer user; (3) the 'adult' child computer user; (4) the 'dangerous' child computer user; (5) the 'victimized' child computer user; and (6) the 'needy' child computer user.

The 'natural' child computer user

The first discourse unproblematically presents the child as a naturally adept user of technology. From this perspective, children are seen to have an *innate* capability to use technology, recently reflected in news stories celebrating the current generation of 'natural-born net babes' and 'net-savvy young' (BBC News, 2000). This discourse has been particularly prominent in political thinking over the last 20 years, with the early 1980s 'keyboard generation' of the early Thatcher era replicated by the 'e-generation' of the early 21st-century New Labour government. In both cases the picture evoked is of the child eagerly engaging with technology of their own accord in a free and easy manner, unprompted and unfettered by adult intervention.

Indeed, the notion of the child as natural user of technology was especially common in the early 1980s when the general adult population was being introduced to information technology for the first time. For example, the *Sunday Times* newspaper (in conjunction with the computer company Commodore) ran an annual 'Young Computer Brain of the Year' competition during the first half of the 1980s to promote youngsters' efforts in 'putting a computer to work' to its readers. Every year the innovative application of microcomputers by readers' children was reported on in

detail, thus demonstrating to its readers 'that there are plenty of computer experts on the way' (*Sunday Times*, 12 December 1982: 40).

In many cases, this discourse of the 'natural child computer users' is based around the juxtaposition of the innately adept child with the less skilled adult – thus offering a model of computer expertise for adults to aspire to; safe in the knowledge that IT is 'easy' enough to be mastered by children. For example:

> Girl of Two Opens Micro Show. . . . A girl of two who as yet cannot read or write yet is believed to be a prodigy with computers opened a four-day computer show in London yesterday. Claire Hirst from Stockport has been playing with a BBC microcomputer for nearly 18 months, helped by her father. She can write simple music and draw graphics. Recently she used a music program to compose her own version of Good King Wenceslas and a drawing program to design her own cards. . . . Organisers of the show said 'what better way is there to let people know how easy it is to handle a computer than to let them see it being operated by a two-year-old'. (*The Times*, 7 December 1984: 3)

In some cases, the image of the natural child computer user as 'good example' is used by both government and private companies as justification for their actions by selflessly claiming to be concentrating their efforts on children rather than adults. Thus computer policies and products are presented as primarily being developed for the younger generation who are naturally interested in such things – providing adults only with the option of 'coming along' for the journey. This notion of 'doing it for the kids' has been embellished over the years in both political and commercial rhetoric:

> All kids are inventors.. . . Its because they're not afraid to get their hands filthy. To eat glue. To use a hammer as a brush. To break something just to see how it works. And to start with the impossible, which is where grownups usually stop. Just a few of the things we're keeping in mind as we invent the new HP. Want to come along? (Marketing material for Hewlett Packard Computer Peripherals, 2001; see Figure 1)

> In all of this [the Thatcher government] was acting partially as a missionary for the new technologies.. . . I particularly wanted to appeal to the young, because they knew more about the new technology than their parents. (Baker [former Conservative Minster for IT and Secretary of State for Education], 1963: 64)

The 'successful' child computer user

Linked to the notion of children naturally being able to use technology are more specific stories relating to the outcomes of their use of IT, especially those relating to the *transformative* capabilities of technology when in children's hands. A common manifestation of this theme is the notion of technology allowing the children to excel at whatever task they have

FIGURE 1



'All kids are inventors'
Source: Advertisement for Hewlett-Packard computer hardware products, March 2001.

applied IT to, transforming an 'ordinary' child into an 'exceptional' child. This discourse is perhaps most familiar and enduring with regard to extending children's learning and general educational performance. For example, throughout the 1980s the printed media, to promote this brave new world of computer-based learning, often indulging in speculation and futurology with headlines such as 'Children will Learn from Computers at Home in the 1980s' (*Daily Telegraph*, 28 September 1979: 8) and 'Brighter Future for the Computer Kids' (*New Scientist*, 18 August 1983: 462).

Children were therefore reported in magazines and newspapers to be freely 'learning through play' (*What Micro?*, 1 April 1984: 103) via 'the electronic Disneyland of computer programming' (*The Times*, 29 March 1983) with the computer being presented to readers as a benevolent and powerful tutor lending 'power to the pupils' (*The Guardian*, 14 January 1988: 15). Moreover, newspapers also featured stories on children's natural ability to use and learn from computers in schools and predicting a new generation of 'Computer Kids: The 21st-century Elite' (*Science Digest*, 1 November 1982: 84). Information technology was therefore highlighted as an area 'Where Pupils Outshine their Teachers' (*The Times*, 6 September 1983: 19) and pupils were now, for example, 'Learning Latin without Tears' (*The Times*, 22 December 1983).

Now, at the beginning of the 21st century remarkably similar stories persist, with ICT being represented as a 'Hi-tech Teacher: Computers in Class Boost Maths and Reading Skills' (*Daily Mail*, 11 July 2000) and computers yet again helping children towards 'Learning to be Latin Lovers' (*Daily Mail*, 18 July 2000). Nor is the transformative potential of the computer just a school-aged phenomenon, as this recent *Sunday Times* report on a 'technology for toddlers' class described its 18-month to 3-year-old pupils:

> He can't hold a pencil yet, but through his lessons on the computer he's learnt to spell his name and knows all his vowels. (*Sunday Times*, 21 July 2001: 70)

In commercial terms, this view of the 'successful child computer user' is also freely replicated in terms of IT allowing children to produce schoolwork far beyond their ordinary capabilities – indeed, often exalting the computer as nothing less than 'magic':

> Now students can present ideas and tell stories in ways they never could before – by making experiences outside of class come vividly to life. . . . Make magic happen. (advertisement for Macintosh I-Mac Computer 2000)

Yet in the political arena, the presentation of such transformative capabilities of ICT is even more breathless. As can be seen from this statement by the then-Minister for Technology, the ability of ICT to transform children's creativity and interaction is often painted in grandiose terms:

> Within the next generation, every book ever written, every symphony ever composed, every film ever made and every painting ever painted will be, with the click of a mouse, within the reach of all children. That is the type of step change which is occurring. I am even tempted to say that the next generation of children will communicate their own symphonies through the new information technologies. (John Battle MP, in Hansard, 1997: col. 1180)

From this perspective, the transformative powers of IT are often extended by politicians and educationalists beyond mere teaching and learning and

into other more problematic areas of education – as in this application of technology to pupil attendance and behaviour when in school:

> In my own constituency Arthur Dye primary school [now] has an Internet terminal, and the head teacher has told me that the project is already having an impact on children's behaviour. Children not only come to school early but they do not want to leave at lunchtime to kick a ball around the field. They also stay on later after school so they can get their hands on the kit . . . (Nigel Jones MP, in Hansard, 1997: col. 1180)

. . . and even highlighting technology's potential to overcome children's learning disabilities, as in the case of this early advertising copy for an Atari home computer:

> An Atari 800 Home Computer and a Father's Love Combined to Help Children Everywhere . . . The story of Fernando Herreda's success began with his son's sight problems. Young Steve Herrera had been born with severe cataracts in both eyes and naturally his father was concerned. Herrera reasoned that the boy's learning abilities could be seriously affected by growing up in a world where he could not see. Having just purchased an ATARI 800 Home Computer, it occurred to Herrera that this could be the perfect tool for testing Steve's vision. So he wrote a program simply displaying the letter E in various sizes. Success! It turned out that Steve could see even the smaller E's without special lenses [and] would play happily with the computer-generated letters for hours . . . Thus 'My First Alphabet' was born, a unique teaching program for children two-years and under. (promotional material for Atari 800, Atari, 1982)

The 'adult' child computer user

If IT is seen to have the ability to transform children's abilities to learn, play and develop as children then a further extension of the 'successful' child computer user discourse can be found in the associated notion of IT transforming the child's abilities into the realm of the adult. In an educational context, a popular manifestation of this discourse has been the reversal of roles between teachers and pupils – with ignorant teachers having to learn from their far more adept students:

> As youngsters of the new generation begin to take IT for granted, they will begin to think that the Internet is passé. They will use it for pragmatic reasons and to get the information they need, and teachers will have to catch up. Teachers will be learning from pupils for once. They will have to learn to switch on and use computers. (John Battle MP, in Hansard, 1997: 1187)

Yet it has not just been in education – traditionally seen as a child-orientated activity – that the child computer user is seen to excel. Another popular variation on this theme is the child computer user as expert business executive. The Conservative government's 'Information Technology Year' campaign in 1982, for example, was based around slogans such

as 'Is your seven-year-old better equipped to run an office than you are?' thus positioning the child computer user as both a threat and a role-model for IT-inept parents. Indeed, a recurring set of news stories throughout the 1980s concerned young 'whizz kids' inadvertently creating successful businesses through their use of computers. Some children were reported to have quickly become expert programmers selling their work to software companies for vast sums of money or, in other cases, computers had allowed children to play stock markets and therefore successfully operate in previously 'adult' arenas, as in the case of this 1982 news story:

> The Software Kid – Online for Fortune . . . The behaviour of 16-year-old David Simons is worrying his parents. He doesn't get into punch-ups at soccer matches or frighten old ladies, but he does spend a lot of time in his bedroom late at night with three companions – all of them computers.. . . In a way, [his father] has only himself to blame. It all began when he brought a personal computer back from the office for a weekend and David caught the bug. In the four years since then David has turned into a computer whizz-kid – and a high-earning whizz-kid at that. When he returns to his comprehensive school this week he'll be £10,000 better off than when the holidays started. (*Sunday Times*, 5 September 1982: 45)

Although revelatory, the tone of these stories is also firmly conciliatory – with children presented as willing to 'show the way' to adults regarding IT. Thus the 'adult' child computer user is rarely, if ever, presented as an overt threat to adult authority – a message exemplified in the case of this child's elevation to a role in a national government on the basis of his computer expertise:

> Jamaican Boy is Government Computer Adviser – A thirteen-year-old Jamaican boy has been appointed to advise the government on the latest trends in computers. . . . His mother said that he had been using computers for ten years and was in a good position to advise the newly appointed technology minister. (BBC News, 1998)

Whereas the majority of these stories concentrate on the child taking up an adult role via their use of technology, a final twist from a commercial perspective is the marketing of the computer's potential to transform *any* child into a successful adult. As this 1987 advertisement for the BBC Master computer argues, the child's progression from a 'first grasp of words' to becoming a 'young tycoon' and 'businessman of the year' is only a purchase away:

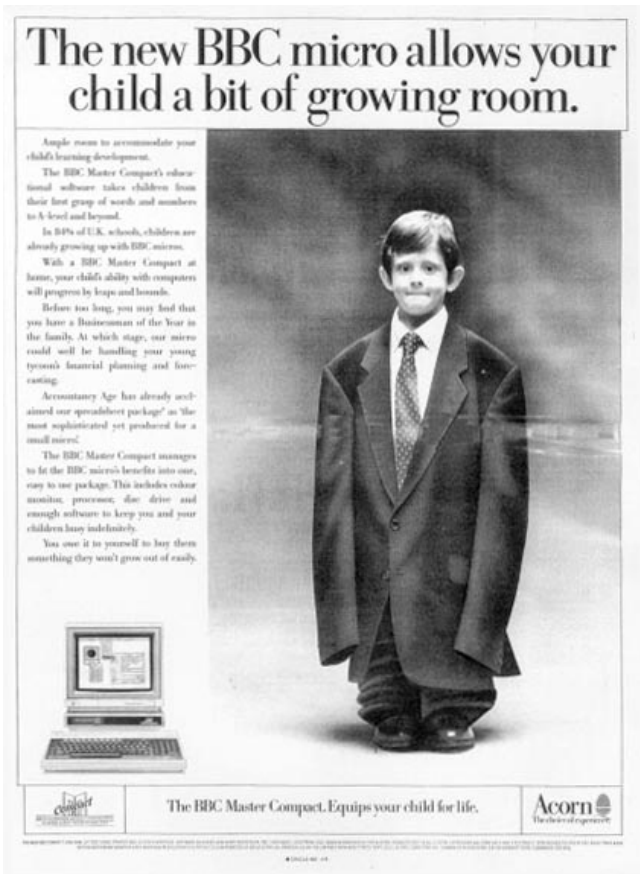
> Ample room to accommodate your child's learning development. The BBC Master Compact's educational software takes children from their first grasp of words and numbers to A-level and beyond. . . . With a BBC Master Compact at home, your child's ability with computers will progress by leaps and bounds. Before too long you may find that you have a Businessman of the Year in the family. At which stage, our micro could well be handling your young tycoon's

> financial forecasting and planning. . . . You owe it to yourself to buy them something they won't grow out of easily. (Acorn advertisement for BBC Master Compact 1987; see Figure 2)

The 'dangerous' child computer user

Whereas the portrayals of the child computer user examined so far have been consistent in their positive and sometimes utopian visions of empowered children using technology for a variety of enriching activities, an alternative dystopian theme also strongly emerges throughout political and popular discourse. One manifestation of this is the notion of what Oswell

FIGURE 2



'The new BBC micro allows your child a bit of growing room'
Source: Acorn advertisement for BBC Master Compact Microcomputer, February 1987.

(1998) refers to as the 'dangerous child'; a child who is actively and aggressively using ICT at the ultimate risk of harming both themselves and others. One strongly recurring theme within this discourse is the deviant child actively seeking out violent and sexual experiences via their computer screens away from the regulation of adults. Thus, as this Conservative MP asserts, such dangerous and undesirable use of technology is seen to take up 'much of children's time':

> Children spend much of their time in the evenings in their bedrooms playing violent computer games or watching computer porn while their parents are downstairs oblivious to the world of their child because they never set foot in the child's bedroom. (Evans MP, Hansard, 1993: 581)

This concern over 'deviant' behaviour and the potential threat to traditional family values was exemplified in the Labour MP George Foulkes's attempts to pass a Control of Space Invaders and Other Electronic Games Bill in 1981. It is testament to the powerful nature of this discourse that Foulkes's Bill was only very narrowly defeated in the House of Commons with an eventual vote of 94 members for and 114 against. In proposing the Bill, Foulkes talked about the 'the force for evil which can arise among young people from addiction to "space invader" machines' and referred to the 'blood money extracted from the weakness of thousands of children'. Most alarming was the assertion that such a transformation was seen as potentially occurring to any child:

> It is interesting but worrying that it is often bright children who have never been in trouble who become hooked on the 'space invaders'. (George Foulkes MP, Hansard, 1981: 288)

Such rhetoric, although overtly alarmist, neatly illustrates the popular conception of the child computer 'addict' – driven to anti-social behaviour by the hypnotic allure of technology. With the computer addict now recast in contemporary discourse as the 'Internet addict' the underlying message of the undesirable side of ICT has remained consistent over the last two decades:

> I have seen reports from all over the country of young people becoming so addicted to these machines that they resort to theft, blackmail and vice to obtain money to satisfy their addiction. . . . This is what is happening to our young people. They play truant, miss meals, and give up other normal activity to play 'space invaders'. They become crazed, with eyes glazed, oblivious to everything around them, as they play the machines. It is difficult to appreciate unless one has seen it for oneself. I suggest that right hon. Members who have not seen it should go incognito to an arcade or café in their own area and see the effect that is having on young people. (George Foulkes MP, Hansard, 1981: 287)

> The terrifying point is that children have an appetite for the most violent of [computer] games, which account for 70% of the market and are being

> promoted with slogans such as 'more fun than killing your neighbour's cat'. . . . In 1998, an advertisement in a children's magazine promoting a game called Vigilance encouraged players aged 13-plus to 'put your violent nature to good use'. The ad was illustrated by a picture of a boy's jeans-clad legs, the barrel of a shotgun at his side and two dead classmates at his feet. The latest games feature rape, torture and mass killing. . . . One in four children who play becomes addicted. Very young children who can't tell reality from fantasy violence are easily hooked. Unlike television, such games engage children's entire attention as they are taken on an emotional rollercoaster that rewards them for killing people. (*Sunday Times*, 28 November 1999: 17)

Throughout all these stories children's natural predilection for engaging with technology is presented as empowering the child in undesirable and anti-social ways. Adults' loss of power to the technology is seen as particularly concerning – especially as children are actively seeking out such experiences and roles as dangerous users of technology. As one media commentator recently argued, 'Now children don't want to be train drivers when they grow up, but hackers' (*The Observer*, 20 February 2000: 19).

The 'victimized' child computer user

The 'dangerous' side of ICT is also reflected in the positioning of the child as an 'innocent' user of technology who may be inadvertently exposed to undesirable violent or sexual material. This subtle shift in emphasis from the notion of the rogue, dangerous child computer user switches attention instead to the 'child-in-danger' from ICT (Oswell, 1998), passive and weak instead of active and aggressive:

> My son is only 12. But with the touch of a few buttons be can unlock a tide of indescribable filth on his computer. The terrible reality is that no one can stop this happening.
>
> One of the *Mail*'s top writers describes how a seemingly harmless educational toy has been turned into a disturbing force for evil. (*Daily Mail*, 15 September 1995)

Unlike our previous examples, this discourse is perhaps the most explicit in focusing on the role of adults in mediating children's use of technology. As the above quotations intimate, at one extreme it is argued that 'no one can stop this happening' and that adults, as well as their children, are powerless in the face of 'evil' technology. Yet, in political and commercial terms, this discourse is more often presented as a stimulus for adult action. For example, the UK government recently launched a series of initiatives aiming to regulate children's use of ICT in a 'safe' and desirable manner – including GridWatch, a variation on the 'neighbourhood watch' scheme where teachers, parents and students are encouraged to report any unsuitable Internet material to the government Department for Education and

Skills. From this perspective, it is the responsibility of adult to prevent the child computer user falling victim to the dangers of the Internet:

> Children should be able to travel along the superhighway in safety, protected from unsuitable material as they tour the web. . . . We share a responsibility to make sure that children's use of the Internet is appropriate and safe. Although much is being done, we cannot afford to be complacent about it. (Wills, 2000: 5)

Moreover, this notion of the child-as-victim of technology is also increasingly being used in commercial portrayals of ICT in urging parents to 'take control' and 'protect' children from the 'hidden dangers' of the Internet:

> Don't ignore the danger online. Take control. Essential Internet protection for your family . . . 33% of UK children with Internet access have found content that has upset or embarrassed them. Much of this is described as rude. Porn, chat rooms and emails from strangers could pose hidden dangers online for children. Help protect your family with Norton Internet Security Family Edition 2001 at only £49.99. (Advertisement for Symantec Software Company, 2001 – see Figure 3)

The 'needy' child computer user

Whereas the first five discourses have been consistent in their portrayal of children's greater technical competence, a final theme concentrates more specifically on children's lack of skills. This discourse is also the most consistent in its construction of children as learners and future adults – a common view of children in society (Moran-Ellis and Cooper, 2000). Thus, the child is presented as 'needy' in terms of gaining the skills needed to use ICT successfully. The onus for this generational skills deficit is therefore placed firmly on educators and parents. For example, over the last 20 years, politicians' exhortations that children should be 'trained' in the use of information technology by schools and teachers have been remarkably persistent:

> The use of computers outside school is not confined to calculation. There is no reason why schools should be any less innovative. We must not forget that when we move on out of this recession we shall need trained young people ready to cope with the latest technology has to offer. (Boyson, 1981)

> [It is vital] to ensure that young people become aware of the value of information technology. . . . Where young people are regularly using technology to enter, use and manipulate information at school, they will be better placed to help industry and commerce to compete effectively. (DTI, 1988 in McNeil, 1991)

> Children cannot be effective in tomorrow's world if they are trained in yesterday's skills. (Blair, in DfEE, 1997)

FIGURE 3



'Don't ignore the danger online'
Source: Advertisement for Norton Internet Security software, March 2001.

From one perspective, this need to help children develop IT skills is argued to be an almost selfless one on behalf of the current generation of adults – preparing children for a technological future that the current generation of adults will not be part of. For example, this was reflected in the motto of the government endorsed and industry-run 1998 'UK NetYear' campaign, 'Connecting Our Children to a Better Future'. Similarly, children must be prepared for the (technological) life ahead of them:

> Children had to learn keyboard skills at school, since whatever they were to do in life they were going to come into contact with the microchip. We have to train the young people of today for the jobs of tomorrow. (Baker, 1993: 61)

Yet this altruistic perspective is also based on training children as future workers to help maintain 'our' economic situation and competitiveness – using children to ensure that 'we' (as British society) do not fall behind:

> Computers, the skills that children will learn and the amount of information they will have at their fingertips will be part and parcel of education throughout the world. If it does not happen here and we do not put computers on our kids' desks as Mr Blair has promised, we will be overtaken in the coming technological revolution by every country from east to west. That is already starting to happen. (John Maxton MP, Hansard, 1996: 483)

> . . . to provide opportunities for all citizens – yes youngsters as well – to get to grips with new technologies. . . . Without the next generation – the children and the youngsters – understanding it, we shall miss out again. (John Battle MP, Hansard, 1997: 1185)

Discussion

As can be seen from these six discursive themes, the portrayal of children and ICT has proved an enduring and multifaceted aspect of the social construction of the 'information age' over the past two decades. Moreover, such discourses, however paradoxical or inconsistent, combine to represent a highly pervasive and persuasive portrayal of information technology as powerful and omnipotent. Given such pervasiveness, it is unsurprising that many people concur with the over-riding idea that younger generations are inherently more skilled and adept at using IT, if sometimes more reckless or vulnerable when doing so. Above all, it is striking how the nature and form of these stories has remained remarkably consistent over the last 20 years when describing children using information technologies from the microcomputer to the Internet.

All these six discursive themes are apparently simple stories about children and technology, but they also consistently reproduce long-established representations of the 'child' as well as earlier stories about technologies such as film, radio and television (see Hendrick, 1990; Wartella and Jennings, 2000). For example, many of the themes are constructed around biologically essentialist views of children as naturally technically skilled yet emotionally immature (Holloway and Valentine, 2001). Thus on the one hand, the perennial construction of children as 'adults in the making rather than children in the state of being' (Brannen and O'Brien, 1995: 730) – or as James et al. (1998) put it, constructing the child as a 'becoming' rather than a 'being' – runs consistently through all the different discourses. The only exception to this rule is the 'adult' child

computer user discourse, the impact of which is based around the power of technology to 'unnaturally' transform the child into a mature, developed state. Yet in all cases, the emotionally immature child is seen as being naturally capable of using ICT. As Buckingham (1998: 557) observes, 'children are seen to possess inherent qualities that are somehow uniquely related to the inherent characteristics of particular communications media'. This unproblematic characterization of the innately adept child contrasts with the Janus-face portrayal of technology capable of both empowering and disempowering the child user – discourses which themselves echo long-established notions of the 'free' child and the 'restrained' child (Jenkins, 1998).

The six different discourses presented in this article also ultimately conform to one of the two dominant paradigms that characterize societal discussion of technology in general. All the stories are underpinned by either a technological or a social determinism, where information technologies are presented as an inevitable consequence of either technological development or the technological needs and requirements of society. As has been pointed out in the recent social studies of childhood literature, such biologically, socially and technologically determinist views of children and technology are obviously flawed. In particular, the over-riding conclusion reached by such researchers is that such discourses of children and technology fundamentally fail to reflect the actual diversity and complexity of real-life ICT use:

> A traditional view of children as essentially innocent and vulnerable to media influence is replaced by an equally sentimental view of them as naturally 'media wise'. Both positions reflect an essentialist construction of 'childhood' and 'youth', and an unduly determinist account of the role of media technology. Both fail to acknowledge the diversity of the lived experience of childhood, and of children's relationships with the media. Neither offers a realistic basis for enabling young people to cope with the changing cultural realities into which they are now born, and in which we hope they will come to participate as active citizens. (Buckingham, 1998: 566)

This is, of course, an important point and has formed the basis of the majority of recent work in the area (i.e. Bingham et al., 1999; Facer et al., 2001; Moran-Ellis and Cooper, 2000; Valentine and Holloway, 1999). Such authors are correct in observing that at a macro level, such flawed discourse undeniably influences policy-making. As Oswell (1998) reminds us 'in falling back on particular existing figures of childhood, [technology] policy could potentially work against the best interests of the child'. Similarly, such stories also have great influence in framing the actions of adult decision-makers such as teachers and parents with regard to facilitating ICT access to young people (Bingham et al., 1999).

Nevertheless, while it is important for academics to challenge such stories for their inaccuracies and self-perpetuating nature, it is also vitally

important to take a wider perspective – one that has been lacking so far – from the child/computer debate. In particular, if we look beyond criticism of such stories as inaccurate and misleading portrayals of children and IT we can, instead, begin to consider the wider societal functions such discourses may serve. This point is not made as a criticism of previous work in this area – merely to highlight a complementary perspective on the same area of inquiry. The emphasis on the child as social actor and 'being' to be understood in its own right which constitutes the epistemological basis of the 'new social studies of childhood' tradition of academic work (Holloway and Valentine, 2000; James et al., 1998) inevitably means that much of recent work into children and technology has shied away from the political-economic basis of such discourses. Yet, it can be forcefully argued that the political-economic function of these stories is key to their enduring existence and effect on society. From this perspective it quickly becomes clear that these discourses are not primarily concerned with children's use of technology but rather function as metaphors for society and technology. In (re)considering the content of these advertisements, political rhetoric and media output we must strive to move beyond solely concentrating on the message itself and also consider the communicator(s) and audience(s).

Thus it can be argued that the six discursive constructions of child computer users identified in this article are all predominantly attempts to 'sell' information technology to the adult population. In this way it is noticeable how all these discourses attempt to enrol readers 'by speaking to or implying a community of responsible readers from which one would be reluctant to exclude oneself' (Moran-Ellis and Cooper, 2000: para. 5.3). By presenting the 'other-worldly' strengths and weaknesses of the computer-using child, key actors in the shaping of the information age are attempting to coerce adults to take action. The fact that this may not be immediately noticeable at least partially stems from the similar teleological assumptions about growth, progression and development which underpin our conceptualization of children, technology and social change in general (Sefton Green, 1999).

Having identified these alternative foundations for the construction of children and technology we should, perhaps, examine further the nature and substance of these stories. If these stories are *not* primarily concerned with children and childhood then *what* or *who* are they about? In doing so, above all it is important to examine what such stories do not tell us about society and technology. As Law (1994: 152) argues, 'all stories, looked at from another point of view, another mode of ordering, are also heroic attempts at suppression'. In particular, the framing of children, adults and technology within these determinist discourses tends to hide the key shaping actors, the values and power relations behind the increasing use of ICT in society. As Lyon (1988: 18) argues, this is a common shortcoming of societal discussions of the 'information society', with popular rhetoric

designed so that it 'obscures vested interests that are involved in IT and that in fact do so much to shape its overall direction . . . yield[ing] no clues as to who wields power'.

The child computer user and the commercial marketing of IT

On the one hand, the portrayal of the child computer user is an obvious element of the marketing of specific computer products, and the wider concept of 'information technology' itself, on the part of IT companies. In all the examples in this article it is noticeable how the messages are strongly directed, in a variety of subtle and not so subtle ways, at adults rather than the children themselves. Although a host of researchers have pointed towards the direct marketing of goods at children as consumers (e.g. Cook, 2000; Montgomery, 2000; Willis, 1991), the various discourses of the child computer user are pitched towards adults, relying on the well-established notion of the child as a consumer, whose needs must be provided for by their parents (Alexander, 1994).

Thus many of the commercial texts examined in the article are directed at adults – urging parents to purchase products which their children will eventually use. This is especially noticeable of the texts from the 1980s, when the child and adolescent market remained the principal target group of UK computer firms (Haddon, 1988). Although the computer as a 'video games' machines quickly took off in popularity at this time, most companies were loath to lose the 'serious' air that had previously surrounded IT and saw education and cognitive development as the best way to position the home computer with a degree of respectability (Haddon, 1988). Thus firms such as Atari and Commodore aggressively marketed their computers as educational tools for children – thus giving a legitimate *justification* for parental purchase whereas games remained the actual *reason* for purchase for many consumers (children if not their parents). As Haddon (1988: 26) observed at the time:

> Eventually many of those companies emphasised the micro as an educational machine. On the one hand this had a wide-ranging meaning: education for all the family. The new term 'edutainment' soon started to appear in the vocabulary of marketers. This captured the idea that home education was also fun and part of leisure activities. The other meaning of education was geared specifically to children. This strategy appealed to parents to think of their children's future, based on the premise that somehow computer skills were the route to future jobs and prosperity. A few years later, the introduction of micros into the education systems of many Western countries was used by manufacturers to support their marketing claims.

Throughout the 1980s the 'Mum and Dad factor' was seen by many IT firms as a vital element in recruiting individuals into the world of

computers (Reed, 2000). Yet many of the advertisements and stories from this time onwards are also subtly selling IT to adults for themselves to use. In this way the child computer user functions as an aspirational icon for the non-computer-using parent/adult; a miniaturized version of the computer-using self if only a computer is purchased. This can be seen in the inference in some of the texts that adults will inevitably become engaged in becoming computer users *through* their children and that some of the IT magic will 'rub off'. With parents writing computer programs for their offspring and teachers having to learn keyboard skills from their pupils, the child computer user is therefore being presented to adults as a conduit to IT-literacy for themselves.

The use of the child computer user to sell IT products to parents is reflected in the generally positive way in which children are presented in these stories. From the portrayal of computer 'whizz kids' in the 1980s to 21st-century school children constructing multimedia school assignments, the general presentation of computer-using children in commercial marketing over the last 20 years has been a positive one. Child computer users are successful scholars, entrepreneurs and playmates, and, even when portrayed as 'dangerous', 'in-danger' or 'needy', are basically decent children given the right technological protection and support. This generally positive portrayal of the child computer user – especially as scholar – in the marketing material of IT firms is in direct contrast with many other commercial depictions of children and young people. In both television and magazine advertising children are seldom portrayed in scholastic roles, and when they have been shown in such roles, children are usually not shown in a favourable light (Peterson, 1994, 1998). As Heiman (2001: 275) argues, in general contemporary media debate and advertising 'practically *all* children have become suspect and pathologized' (her emphasis). Yet the child computer user continues to be portrayed as a more-or-less successful scholar and social being – thus further emphasizing the unique transformative potential of IT for both children and adults.

The child computer user and the political justification for IT policy-making

The notion of the child computer user has also functioned as a powerful means of 'selling' government strategies and policies to the general public behind the premise of meeting the needs and requirements of future generations of IT users and, it follows, the nation. Indeed, the discursive combination of children and technology – particularly the determinist notion that children must adapt to the relentless technological changes in order to survive – has been used to great effect over the last two decades

by governments of all political persuasions to render IT policies almost unopposable. As Lemke (1995: 58) asserts:

> . . . the political advantage of this technocratic strategy to those who practice it is that it presents policy as if it were directly dictated by matters of fact and deflects consideration of *values* choices and the social, moral and political responsibility for such choices.

Thus, over the last 20 years, there has been a thoroughly bipartisan approach to technology policy-making in the UK with little or no political disagreement. As Adamson and Kennedy (1986) wistfully observed at the height of the Conservative government's IT drive, unlike most areas of life 'it was difficult for the rest of the world to ignore or disapprove of the computing obsession'.

Of course governments and political parties have long used children as a means for selling political ideologies and policies – from the symbolic use of children in presidential campaigns (Sherr, 1999) to the recent use of a young girl to promote the single European currency to the French population. The use of the child as a political symbol has therefore extended easily to the promotion of technology. Yet, in the case of IT, children have been strongly identified as being both the reason for and the recipients of these policies. For example, as Oswell (1998: 278) has ably demonstrated, in much official documentation 'the child is imagined, and childhood is figured, as a particular problem for regulation' and legislation. Similarly, in education, children's future need for 'computer literacy' and other ICT skills is often presented as sufficient reason for the 'technologizing' of education (Selwyn, 2000).

Of course governments have also attempted to use IT to sell policies and initiatives to the children themselves – particularly in the case of education. As the then-Secretary of State for Education eulogized, 'using computers has status within youth culture, where books too often have not' (Blunkett quoted in McGavin, 1997). Whereas their parents may have been seduced by such rhetoric, attempts to sell both technology and education back to children in this way have proved far less of a success – perhaps underlining the naivety and over-simplifications of the view of children's relationships with technology that the child computer user discourse relies on. A succession of educational computing initiatives throughout the 1980s and 1990s failed to make a sustained and lasting impression on UK school children (Selwyn, 1999b) and, despite the convictions of politicians and policy-makers, the allure of IT alone looks unlikely to persuade swathes of disaffected and 'disappeared' youth to return to schools and engage in formal education as present politicians may believe. Here, then, is one example of the selling power of the child computer user discourse on the wane – ironically, among the children themselves.

The child computer user and the wider 'selling' of the information age

Although the child computer user has both these obvious commercial and party-political elements there is also a broader underlying symbolism at work – the use of the child computer user as means through which to rehearse societal anxieties and crystallize our shared conceptualization of the 'information society'. Robins and Webster (1989) point towards the construction of a prevailing 'moral climate' throughout the 1980s regarding the societal desirability of IT and the same can be said of the 1990s and the Internet. Certainly, through media, government rhetoric and commercial marketing over the last 20 years there has been a concerted, continuous effort to forewarn and prepare citizens for the imminent lurch away from industrial towards post-industrial society. Viewed in this light, much of the 'child computer user' discourse can be seen as an integral part of the promotion of the information society as an economic, political, cultural and societal 'matter-of-fact'.

Indeed, much of the governmental and commercial construction of the child computer user makes more sense when contextualized within the political and economic needs of the time. For example, during the 1980s, the political impetus towards IT can be firmly positioned within the context of Britain's economic decline over the 1970s and early 1980s and the subsequent concern with the so-called 'British Disease'. In the midst of rising levels of unemployment and perceived British failure to adapt to industrial change, information technology was furiously seized upon by politicians as a high-profile antidote to the economic difficulties of the times (Robins and Webster, 1989). Allied to the view of the 'British Disease' in industry were concurrent concerns about the state of UK education – initiated by Prime Minister Callaghan's 1976 Ruskin speech and the subsequent 'Great Debate' about education. Thus the government's concern throughout the 1980s of establishing a national 'frame of mind' at ease with, and supportive of, the IT revolution adopted a strongly economic and educational approach. This approach was firmly based around the ideologizing of 'information technology' and the British IT industry – and then making these notions concrete via a programme of education and training reform for preparing upcoming generations of computer users. As McNeil concluded, this 'peculiarly British' approach made children and education the obvious choice to 'carry the burden for Britain's place in the international IT race' (1991: 133). Similarly, from the early 1990s onwards, political rhetoric and policies dealing with children and IT have gloomily reflected the uncertainties of globalized economic activity and the information-dominated 'knowledge economy'. The child computer user, it seems is a convenient means through which to express and exercise these wider concerns.

Societal concerns over the economic future of the looming post-industrial society are therefore reflected in the marketing of IT through images of children as the computer-using businessman of tomorrow, as well as other positive evocations of child computer users as miniature and nascent IT-workers. In the political and popular cultural representations of the child computer user during the 1980s, there are, for example, hints of an interesting tension between the political insecurities as to the potential continuation of the 'British Disease' in the information age and the financially confident and ambitious 'Yuppie' work culture of the time. Children were therefore being used to exemplify and model the utopian and dystopian post-industrial futures that the politicians and business-people of the 1980s imagined could be in store. Similarly, it could be argued, as the 1990s progressed and the more mundane realities of IT-based employment began to become realized, then the notion of the child computer user gradually refined to encompass new uncertainties about the technological future, such as concerns over globalized computer networks and the need for flexible work patterns and attitudes as well as technological multi-skilling.

Thus, throughout the 1980s and 1990s, the child computer user has continually been used as a discursive space to exercise anxieties of the near-future of the information age, and the production of desires and dispositions of people as workers, citizens and consumers in post-industrial society. Talk of the computer as a transformative tool for children can therefore be seen as a metaphor for the computer as a transformative tool for economies, businesses and adult employment. Through the continued metaphorical use of the child computer user as computer-using worker, governments, businesses and the media are attempting to orient adults in, and familiarize them with, the new capitalism of the information age. In one sense the child computer user is therefore being used to (re)affirm the new required positions for people within the relations of production within post-industrial capitalist society. For example, in the 1980s we were strongly being reminded through images of our children (and therefore ourselves) that the new economy would need computer programmers, IT-using businessmen and a computer-literate population of workers in order to 'compete' and 'win'. Now, in the early 21st century, we are being told of the need to be Internet-savvy and technologically flexible. In both cases the generational nature of the child computer user serves to highlight the competitive nature of this transformation – 'You had better adjust to this new way of working because your children certainly will!'

Yet the child computer user should be seen as constituting more than merely political and popular concerns over future workforces and economies. There are also strong societal elements to these stories as can be seen, for example, in recurring concerns over computers and the Internet constituting new social spaces with uncertain boundaries and rules. The

themes of the 'dangerous' and 'victimized' child computer user typify these societal concerns over the uncertainties surrounding the regulation and use of computer-mediated environments such as video games and the Internet – thereby functioning as both cautionary and celebratory stories. In all these cases, the key point is that the child computer user discourse continues to be addressed as much to adults as it is towards children. As issues of regulation, surveillance and 'appropriate' use have emerged as key societal concerns over technologies such as the Internet, so these themes continue to be reflected and enforced in portrayals of the child computer user.

Conclusion

The crux of this argument is a simple one – that the notion of the child computer user has been used by state and commercial interests to rehearse, clarify and promote a range of societal, economic and political concerns over the role of new technologies in society. Yet it offers an alternative perspective to suggestions that portrayals of children and technology in political and popular culture are misleading, inaccurate and should therefore be dismissed or ignored by serious observers of technology and society. Instead it can be argued the notion of the child computer user, is to a large degree, merely a means of persuasion and promotion on the part of the key commercial and political guiding interests of the information age:

> Beyond the control of resources, capitalists must be able to create legitimacy for themselves in their role of resource developers, and this requires the capture of public discourse. When the captains of industry are not capable of responding to people's real needs, it becomes necessary for them to change the popular perception and language of 'need'. (Sussman, 1997: 30)

Thus, as we have seen, the notion of the child computer user is as enduring as the notion of the information society itself. But in assessing the significance and effect of these stories we need to be aware of what these discourses of the child computer user include and exclude. In doing so, a host of questions come to the fore: Who is and who is not a computer user? What is and what is not the role of the private sector? What is the role of state apparatus such as the media and government in shaping the information age? What is and what is not the role and capabilities of the technology itself? Even briefly reconsidering the six discursive themes with these questions in mind, we can see that the child computer user is more white, male, non-disabled and from the 'professional-managerial class' than not. Indeed, alongside many other cultural portrayals of children, the child computer user is presented and evaluated from a typically Western, middle-class perspective (cf. Ponte, 2001). Although

such portrayals may well be less gendered now than maybe was the case in 1981, there are still many subtle contradictory messages about the roles and capabilities of boys and girls, black and white, 'rich' and 'poor' that need to be explored. As it stands we are being sold a very economically and politically narrow picture of the information age where the solution to all our problems simply lies in the consumption and use of technology.

Such stories also divert attention from the key actors standing to gain from societal adoption of ICT – in other words the real 'winners' of the information age. As Sussman (1997: 25) identifies, the 'technocracy' within government and business may be telling these stories, but their role is conveniently obscured by discursive devices such as the child computer user:

> Technological determinist language . . . gives the impression that technological developments unfold as inevitable progressions, driven by their own momentum, as if economic conquest and expansion and the ruthless appropriations of property and wealth are irrelevant considerations. The political value of such language allows the technocracy within business and government to present itself as proceeding without passion or favouritism; and, above all, with the neutrality toward partisan interests that leaves it free to select and implement the best technological solution.

Taking this standpoint leads us neatly into framing the key questions that *should* be asked of children and adults in the information age. Only by challenging the very notion of the information age as it is presented to us via the child computer user discourse can we begin to uncover the economic, political, cultural and social issues that underpin the fortunes of individuals. Of course the information society is a nebulous concept – at best referring to a host of economic, social and cultural changes seen to be driven or at least shaped around the ongoing development of information and communications technology (Lyon, 1996; Selwyn and Gorard, 2002). Yet, as Robins and Webster (1999: 89) contend, 'in this cocktail of scientific aspiration and commercial hype, there are a number of implicit but significant assumptions that need to be challenged throughout any analysis of technology and society. For example, to what extent are our economic fortunes and those of our children dependent on the 'information sector'? What actual demand will there be for 'information workers' and what will such jobs actually entail? What is the purpose, function and content of 'information work'? Is IT leading to the expansion and extension of opportunities to all social groups? How is access to technology patterned according to individual factors such as age, gender, class, geography (both in terms of distance and terrain) and ethnicity, and most importantly, how are different social groups then able to make use of this access? Most crucially, we need to examine the revolutionary nature of the information age – how 'new' and 'different' are these changes? To

what extent does IT usher us into a new kind of society for both adults and children? It is to these questions that the debate on children and technology should now turn.

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